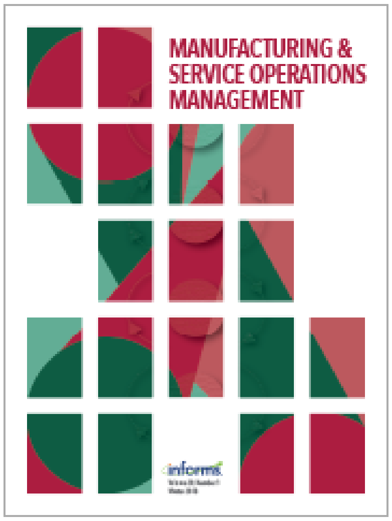

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The Impact of Digitization on Print Book Sales: Analysis Using Genre Exposure Heterogeneity

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Abstract. ***Problem definition*:** The rise of digital channels has led to significant media market transformations. This paper studies whether and how digitization, sparked by the launch of Amazon Kindle in late 2007, affected print book sales. ***Methodology/results:*** To estimate the impact, we exploit the quasiexperimental variation in the popularity of digital books across different genres or subgenres. We employ difference-in-differences and other identification strategies, and we use print sales data on a large representative sample of book titles published in the United States from 2004 to 2015 across a variety of genres. Using various empirical specifications, we find that digitization significantly reduced print sales of adult fiction (the most popular genre in the e-book format) while having a much smaller impact on adult nonfiction and juvenile fiction. We further find that the effect for adult fiction is higher after the launch of the iPad (in 2010) and stronger for smaller publishers, the paperback version, and low-selling books. ***Managerial implications*:** Our findings provide book publishers with key inputs for optimal pricing and release strategies of different book formats. Our study also extends the vast yet growing academic literature on the impact of digital distribution channels on existing sales channels and can inform the policy debate concerning third-party digitization and its effect on physical sales.

Supplemental Material: The online appendix is available at https://doi.org/10.1287/msom.2022.0594.

Keywords: e-book • cannibalization • genre heterogeneity • e-readers • books

1. Introduction

History has taught us that market disruptions created by the introduction of new technologies have uncertain effects on the demand and production of existing goods. Take, for example, how the movie industry during the 1980s initially feared the disruption caused by the introduction of video recording, which led to a slim Supreme Court 5–4 majority in favor of permitting the making of individual copies under fair use (Sony Corp. of America v. Universal City Studios, Inc., 464 U.S. 417 (1984)), and how it later on embraced video recording as a major way to reach customers. Similarly, although record companies initially feared cannibalization from radio stations in the 1920s and attempted to secure royalties for playing their songs, they eventually sought to make payments to radio stations after learning that radio airplay increased record sales (Sterling and Kittross 2001). More recently, the rise of digital channels has led to significant disruptions in many media markets, which led to a broad debate on whether new forms of delivery of information goods complement or cannibalize traditional forms of delivery and how these digital disruptions will transform these markets (e.g., Gentzkow 2007, Smith and Telang 2009, Liebowitz and Zentner 2012).

This paper focuses on how the disruption caused by the digital delivery of books affects the book publishing industry. In this industry, publishers face the challenge of selling books in both print and digital formats. Since the launch of the Kindle e-reader in late 2007, the industry has witnessed remarkable growth in both the adoption of e-reading devices (e.g., e-readers and tablet computers) and digital book sales. According to a survey report from Pew Research (2015), the adoption of e-readers or tablets has been steeply increasing since 2008, with more than 45% U.S. adults owning at least one e-reading device in 2015 (see Figure A.1 in the Online Appendix). Relatedly, PwC reported that e-books generated over $4.5 billion in revenue in 2013, up from just $270 million in 2008 (Statista 2014). Similarly, according to Nielsen, the digital format accounted for 24% of the total unit sales in 2015 compared with merely 8% in 2010 (Milliot 2016). This huge surge in e-book sales thus raises a natural question of how e-books affect sales in print format.

There has been an active debate among industry experts on whether and to what extent e-books substitute or complement print books. On the one hand, as content is the same in both formats, the availability of the e-book version may cannibalize print sales, especially because e-books are generally priced much lower than print books. Consumers could also have strong preferences for books in digital format, and thus, the demand for print books could decrease because of a combination of both price and format preferences. Concerned by the potential cannibalization of print sales, some publishers decided to delay the release of their titles in e-book format (Rich 2009). Publishers also sought to suppress substitution by controlling e-book prices, which gave publishers the ability to charge higher prices for e-books until this practice was legally challenged (De los Santos and Wildenbeest 2017).

On the other hand, however, several industry experts, including Amazon's Chief Executive Officer Jeff Bezos, and anecdotal evidence (Janas 2017) suggest that e-books do not cannibalize print sales (Kaplan 2009). It is argued that most buyers of books in physical format are loyal to the physical channel and thus, do not decrease their purchases of physical books. Under this view, sales in digital format only represent incremental sales that would not have occurred if e-books were not available. Moreover, buyers of e-books help in a product's discovery (e.g., via word of mouth), which in turn, increases sales through the physical channel (Peukert et al. 2017, Nagaraj and Reimers 2023).

The nascent literature on the impact of the digital channel on physical book sales provides, at first blush, starkly contrasting answers to the question of whether e-books substitute print books. Chen et al. (2019) compare sales across books, exploiting a change in the release strategy by a publisher that during some weeks in 2010, delayed the release of the e-book version of 99 titles but during other weeks in 2010, released 83 titles simultaneously in print and e-book format. They interpret their findings as suggesting no evidence of a significant impact of such a delay on hardcover book sales in any genre. Their interpretation would suggest that e-books do not substitute print books. However, Li (2021) estimates a structural model using a sample of about 73,000 online transactions of books and county-level physical bookstore sales data from surveys and projections between 2008 and 2012, and she interprets her findings as suggesting that 72% of e-book sales come from cannibalizing print sales. Thus, her interpretation would suggest that e-books substitute print books. Thus, the interpretations of Li (2021) of her results are in direct contrast with how Chen et al. (2019) interpret their results.

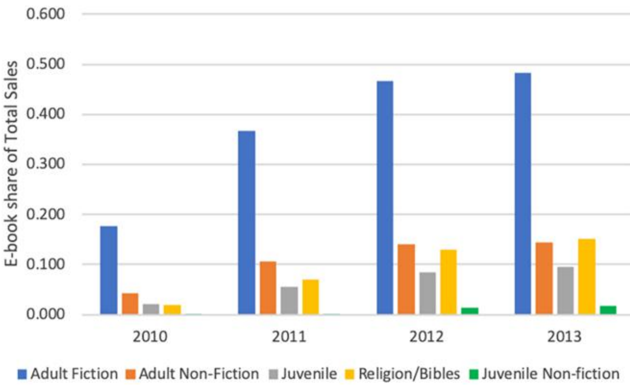
It is worth noting, however, that the empirical evidence from both previous studies on the topic use is seriously limited for various reasons. For instance, the county-level data used by Li (2021) employ data from a company that makes projections based on assumptions rather than actual measures of physical bookstore sales (see Section 2 for more details). As for Chen et al. (2019), they use a very small sample of books in only the hardcover format from 2010 when e-books were in their infancy.

In this paper, we seek to estimate the impact on print book sales because of the rise in digitization, led most notably by the launch of the Kindle e-reader in late 2007 and the iPad in 2010 but embracing a variety of demand- and supply-side factors that are often considered as part of "digitization."[1] To do so, we analyze yearly print sales data on a very large representative sample of book titles published and sold in the United States between 2004 and 2016 obtained from Nielsen/NPD, which are considered the gold standard for data on book sales. We employ a difference-in-differences (DiD) identification strategy by exploiting the quasiexperimental variation in how e-book readers are expected to affect different genres and subgenres, such as in Peukert and Reimers (2022), and also employ an alternative identification strategy using the evolution of the share of e-book sales by genre. Genre variation has also been used for identification in other contexts; see the work by Zentner (2005), who examines how file sharing affected music sales in different genres.

In our context, e-readers are most suitable for recreational and leisure reading because of their portability and because of the straight narrative and sequential reading of fiction books as compared with nonfiction (Budiu 2014). This suggests that, in theory, the introduction of e-readers likely provides the most utility to readers of the adult fiction (AF) genre and would have the strongest impact on adult fiction books. On the contrary, few children owned e-reading devices or used electronic devices to read books during our study period, which would make juvenile (i.e., ages 0–11) nonfiction one of the least affected genres (Nusca 2009, FCG Media 2010, Reddit 2023). Furthermore, when reading to children, parents strongly prefer (at least during our study period) to read from physical books (Richtel and Bosman 2011, Pew Research 2012). Similarly, survey data indicate that e-readers are not as well suited for reading textbooks within the nonfiction category (Foasberg 2011).

Data on e-book popularity across genres confirm these theories, complementing the evidence from the surveys mentioned above. In Figure 1, we plot the e-book share of total unit sales from 2010–2013 for different categories of books. As one can see, there is a significant variation in the popularity of e-books across genres, with adult fiction books being far more popular in the digital format than adult nonfiction (AND), juvenile, or religion (REL) books. The e-book share of total sales indicates the level of "treatment" for a genre. See

Figure 1. (Color online) E-Book Share of Total Yearly Unit Sales by Genre



Source. Adapted from Nowell (2015).

Section 3 for more details. Specifically, e-books accounted for about 38% of total unit sales in adult fiction on average across the four years (2010–2013) compared with only about 11%, 6%, and 9% in adult nonfiction, juvenile, and religion, respectively.

Because theory and survey data suggest that e-readers are more suitable for leisure reading and this is confirmed by the data that show that the digital format takes a large share of total adult fiction sales, we use adult fiction as our main treatment group, although we also present results examining how digitization affected sales of books of other genres. It is worth noting that the adult fiction genre is one of the biggest genres in the book industry, generating over $6 billion in 2023 (Milliot 2024). We use several genres (e.g., juvenile fiction (JF) and adult nonfiction (AND)) and subgenres (e.g., religion) as control groups. Among the various genres that we use as the control group, we believe that the use of juvenile nonfiction (JNF) has some advantages. (i) Few children ages 0–11 owned e-reading devices or read e-books during our study period, resulting in the lowest digital format popularity (see Figure 1), and (ii) it offers the best representation of counterfactual trends for adult fiction print sales (we show below that adult fiction and juvenile nonfiction have parallel pretreatment trends).

Unlike Chen et al. (2019), who find a statistically insignificant impact on print sales because of a delay in the release of e-books of any genre in 2010, we find that digitization substantially decreased print sales of adult fiction books by about 30%. We also find that the effect size for adult fiction varies over time, with little effect between 2008 and 2011 and a much larger impact afterward coinciding with the development of the e-book industry. Furthermore, although it is not straightforward to compare our results with the results in Li (2021) and unclear whether the comparison is valid because the data used in Li (2021) do not separate e-book from print book sales and are projected (as we explain in Section 2), our estimates may suggest a lower cannibalization rate than that resulting from the reported estimates of Li (2021).

We perform several robustness checks, such as using different control groups, accounting for supply-side responses to digitization (like differences in the supply of books among the genres), accounting for the rise in self-publishing (perhaps leading to a lower quality of books after digitization), employing an alternative empirical identification strategy using the variation in the share of e-book sales by genre, and using propensity score matching. We find fairly consistent results throughout these robustness checks, with estimates for the effect on print sales of adult fiction books ranging from 24% to 38%. We also rule out potential demand-side factors, like consumers switching to other forms of internet media, that could have possibly explained our results. Furthermore, we also estimate the effects for genres other than adult fiction (e.g., adult nonfiction and juvenile fiction) and find that the effects are smaller and statistically insignificant.

Our study complements the empirical work studying the effect of e-books on print sales (Chen et al. 2019, Li 2021). First, we analyze print sales data over a much broader time frame (2004–2016), covering a time period when the use of e-books had matured, which allows us to observe book sales (of both paperback and hardcover formats) not only before the launch of the Kindle e-reader in late 2007 but also, after 2013 when the adoption of tablets and e-readers had relatively matured. Second, our data are much more comprehensive in terms of representation of books and also, their sales, covering a large sample of the book titles that were published and sold in the print format both online and in physical stores in the United States between 2004 and 2015 across a variety of genres. This not only enables our identification but also lets us estimate the impact for the genres that are most likely to be affected by e-books with enough statistical power. We further extend this literature by analyzing how the effect of the e-book channel on print sales is moderated by publisher size, print format (paperback/hardcover), and the level of book popularity (best selling/low selling).

2. Related Literature

Our paper is related to the literature studying cannibalization between physical and digital channels (e.g., Gentzkow 2007, Smith and Telang 2009) and to the studies on the book industry (Chevalier and Goolsbee 2009).

One strand of the literature on crosschannel cannibalization investigates whether and how the addition of the internet channel cannibalizes sales in established offline channels. Gentzkow (2007) finds that the introduction of online newspapers resulted in a significant decrease in print readership. Liebowitz and Zentner

(2012) find that the internet's negative effect on television viewing decreases drastically with age, ultimately having no impact on the oldest Americans. More recently, there are also studies that examine the opposite by studying how the opening or closing of an offline retail store affects sales in the growing omnichannel retailing literature (Verhoef et al. 2015, Soysal et al. 2019). A general finding in this literature is that the introduction of an offline retail store typically increases online sales (Bell et al. 2018, Kumar et al. 2019).

In the context of information goods, there is a growing literature on how digital distribution channels influence sales in existing channels. The impact of digital piracy or antipiracy laws on the physical sales of music and movies has been extensively studied (Zentner 2006, Oberholzer-Gee and Strumpf 2007). Danaher et al. (2010) examine whether legitimate digital distribution channels, such as Apple's iTunes Store, impact piracy and also, physical sales. They show that the removal of legal video content on iTunes is associated with a significant increase in the demand for pirated content but has no statistical impact on physical DVD sales.

Although crosschannel effects have been extensively studied in the context of movies, music, and television shows, they have received limited attention in the context of the book publishing industry. Kannan et al. (2009) study the optimal pricing strategies of the National Academies Press, which sells both print and digital books, whereas Hao and Fan (2014) theoretically analyze the pricing of e-books and e-readers under the wholesale and agency pricing models. Nagaraj and Reimers (2023) use evidence from the Google Books Project to estimate how free digital distribution affected the demand for physical works. However, their work analyzes very old and nonpopular titles, whereas our book sample is more recent and includes many popular books. Also, our paper is about an alternate channel for reading, whereas Google Books is more about search and discovery. In a closely related work, Chen et al. (2019) use a change in the release strategy by a publisher in 2010 and data from a small sample of 182 book titles in the hardcover format to find that delaying the release of e-books by a few months did not lead to any increase in hardcover book sales of any genre, although it significantly reduced digital sales. Although their interpretation of their own results is that e-books do not displace print books, we believe large standard errors (likely because of a small sample and the use of the year 2010 when the e-book market was very small) cannot be interpreted as a real zero.

In another related paper, Li (2021) estimates a structural model to simulate counterfactuals and finds that 72% of e-book sales come from cannibalizing print sales between 2008 and 2012. However, the data used for estimation are marred by selection and measurement issues, making the analyses unreliable. First, Li (2021) uses a sample of 72,619 online book transactions of books from 20,637 households of the Comscore panel, which changes every year and excludes transactions made from physical stores. We note that sales of hardcover books accounted for only 5% of the online transactions in Li (2021), although they account for 30% in our data, covering a large fraction of the population of online and offline sales. Second, the county-level data on the number of physical bookstores' print book sales come from the 2010 Esri database. Not only are the data for other years imputed using the Esri data and other data sources, but it is also important to note that according to Esri, they do not collect data, but their county-level data itself come from projections based on the Department of Labor's Consumer Expenditure Survey data and other data sources. However, the data on book sales from the Consumer Expenditure Survey are available only at the level of the metropolitan service area (MSA) that includes multiple counties, and even then, it is available for only a handful of MSAs. Thus, it is unclear how Esri "projects" estimates at the county level for every county.[2] Moreover, the data from the Consumer Expenditure Survey do not separate e-book from print purchases, and then, it is unclear how the dependent variable can measure print sales excluding e-book sales.[3] In addition, the model in Li (2021) assumes that consumers who do not own a Kindle e-reader derive no utility from e-books. This is a strong assumption as (i) the Kindle app has been available on smartphones since 2009 (Indiviglio 2010), and (ii) a large proportion of users read e-books on devices other than the Kindle e-reader (Pew Research 2012). Furthermore, unlike our paper, the demand-side model of Li (2021) also does not take into account any potential changes in genre preferences over time.

The current work addresses the shortcomings of the past work and complements this prior work in several important ways. First, we analyze book sales data from a much broader time frame (2004–2016), which allows us to observe book sales not only before 2007 when the Kindle e-reader was introduced but also, after 2013 when e-book sales had stabilized. Second, our data on over a million book titles that were published between 2004 and 2015 across a wide variety of genres cover a large sample of the books published in the print format in the United States during this period. This crucially allows us to estimate the impact on adult fiction (the most popular genre in the e-book format) and also helps in providing us with proper control groups for our DiD approach. Further, unlike Chen et al. (2019), who focus on the hardcover format exclusively, we are also able to analyze the sales of the paperback format. Unlike Li (2021), who focuses on a sample of online book sales and attempts to use highly aggregated projected data on physical bookstore sales, although these data do not allow for separating physical versus e-book purchases, we are able to analyze sales of print books both online and at physical stores at the title level. Moreover, unlike both papers, our comprehensive data allow us to study the heterogeneous effects of digitization on print sales across different years, publishers of different sizes, books in different formats, and books of varying popularity levels. Table A.1 in the Online Appendix presents a summary of a selection of empirical studies related to the interaction between the physical and digital channels across different industries. One can note that the results in this literature suggest that the introduction of one channel does not always cannibalize and may sometimes increase the sales from the other channel.

Last and importantly, our results that digitization reduced print sales in adult fiction are in direct contrast to how Chen et al. (2019) interpret their results, possibly because of a number of reasons. First, they analyze data from 2010 when e-book adoption was very low (see Figure A.1 in the Online Appendix). Second, they considered the effect on sales of only hardcover books, which because of their higher price point, are likely a poorer substitute for e-books than paperback books. Another possible explanation of the different conclusions in our paper versus those in Chen et al. (2019) is that the statistically insignificant results in Chen et al. (2019) might be because of the small sample of 182 books that they use. Although it is hard to directly compare our results with those in Li (2021) because of very different data and approaches, we discuss below that our estimates for the period 2008–2012 (that overlaps with Li 2021) indicate that about 48% of adult fiction e-book sales came from cannibalizing the print format. This is substantially different from the cannibalization rate of about 72% found in Li (2021) for all of the genres together; note that digitization had the most impact on the adult fiction genre. Because the data in Li (2021) originate from data from the U.S. Census that do not separate e-book from print book sales, it is unclear whether the comparison is actually valid.

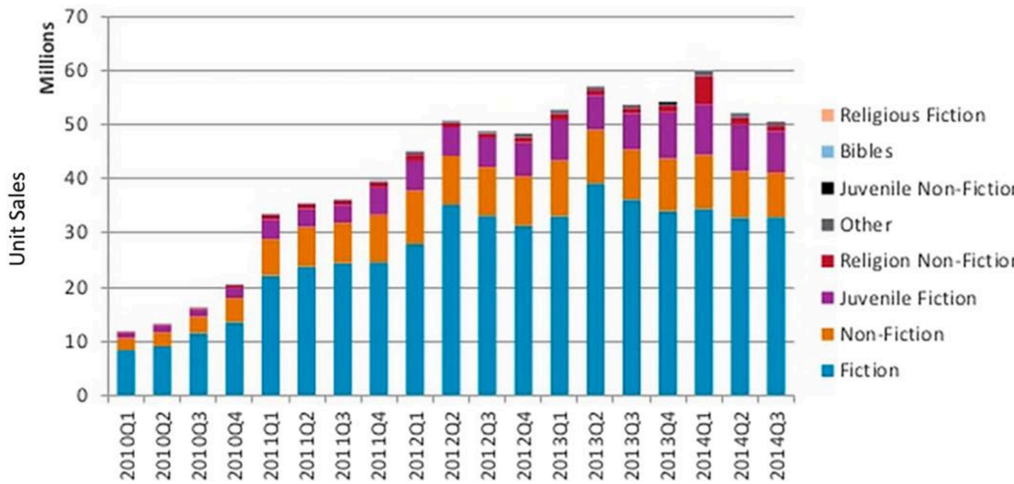
3. Identification Strategy

In order to estimate the impact of digitization on print book sales, one could ideally design a randomized experiment in which the treatment group of books is published in both digital and print formats, whereas the control group of books is only published in print format. Alternatively, it would have been ideal to have access to individual-level data and exploit any exogenous variation in e-reader adoption across individuals. In the absence of such randomization or exogenous variation, we exploit the quasiexperimental variation in how various genres are exposed to e-book shock. Although every book is treated in a theoretical sense, the exposure to the e-book shock significantly varies across different genres of books because of their inherent characteristics; and we exploit this variation to identify treatment and control groups for our DiD approach, yielding an average treatment effect on the treated. Peukert and Reimers (2022) use a similar identification strategy to study a different question. Also, in cases where an entire population is virtually treated, using a continuous instead of a dichotomous treatment is a long-established empirical strategy (see Duflo 2001, Liebowitz and Zentner 2012, Danaher et al. 2020, Peukert and Reimers 2022). For example, many papers use the variation in the proportion of internet use across geographical locations to examine how internet use and piracy affected television viewing, music sales, or newspaper sales. Geographical locations with a higher proportion of internet use or piracy are exposed to a larger treatment than geographical locations with a lower proportion of internet use. Furthermore, it is common to create control and treatment groups based on very low and high levels of treatment, respectively, across different cross-sectional observations. For instance, Peukert and Reimers (2022) use the genre with the highest popularity of digital self-published works as the treatment group and genres with lower popularity as the control group. Similarly, Chetty et al. (2013) use areas with little knowledge about a policy as the control group, and Kummer et al. (2020) use districts with a large (small) change in the unemployment rate as the treated (control) group.

In theory, e-books are more well suited for reading recreational or leisure titles, such as adult fiction titles, because of their straight narrative and sequential read as compared with nonfiction titles (Budiu 2014). Thus, it would mean that in theory, the introduction of e-readers or tablets likely provides the highest utility to readers of the adult fiction genre and would potentially affect adult fiction books the most. On the other hand, few children owned e-readers or read books on digital devices during our study period (Nusca 2009, FCG Media 2010). This can also be confirmed empirically through Figure 1 shown before, in which we see very high (low) sales of e-books in adult fiction (juvenile nonfiction). In fact, e-books accounted for less than 1% of the total unit sales of juvenile nonfiction books in 2016 (Milliot 2017). Further, in Figure 2, we present quarterly e-book unit sales by genre from the first quarter of 2010 to the third quarter of 2014. Again, we observe that, on average, AF accounts for about 65% of e-book sales across all genres. On the other extreme, the share of JNF e-book sales as a percentage of all e-book sales across all genres is negligible.

Note that a large fraction of e-book sales (e-book sales/(e-book sales + print sales)), for the adult fiction genre after the introduction of e-readers does not imply that absolute levels of adult fiction print sales have decreased (or increased) compared with the period before the introduction of the digital readers. The larger

Figure 2. (Color online) Quarterly Unit E-Book Sales by Genre



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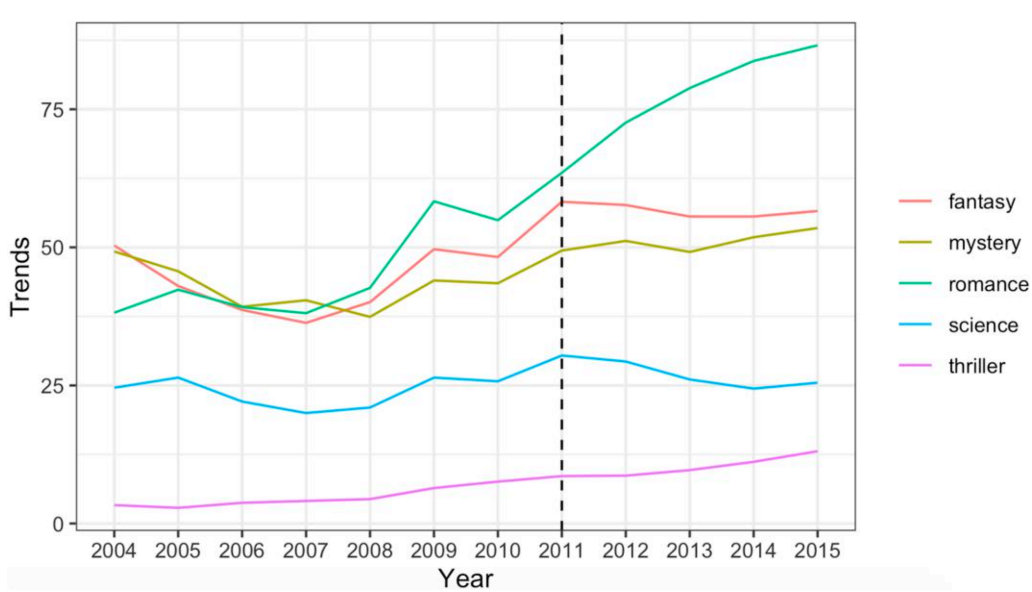
fraction is only indicative of the higher popularity of adult fiction books in the digital format than that of juvenile nonfiction books in the posttreatment period, and it does not inform anything about the changes in print sales since the launch of e-readers.[4]

In light of this, we use adult fiction books (the most popular genre in e-book format) as our preferred treatment group and juvenile books (the least popular genre in e-book format) as our preferred control group, although we also use other genres and subgenres as treatment and control groups as well (e.g., Duflo 2001, Chetty et al. 2013, Danaher et al. 2020, Peukert and Reimers 2022). We also employ an alternative empirical approach using the evolution of the share of e-book sales by genre in Section 5.5. A caveat regarding our identification strategy is that it could be challenged if an unobservable coincides with the timing of digitization and that also affects adult fiction print sales negatively (positively) or juvenile nonfiction print sales positively (negatively).

We note that the release of one of the all-time bestsellers, *Fifty Shades of Grey* (FSG) in 2011, led to a huge positive shock in the demand for romance books that form a major part of adult fiction books (Bosman 2014). This change in preferences can be clearly seen in Figure 3, where we plot Google search trends for terms related to books from popular subgenres of adult fiction (e.g., "romance books," "mystery books," etc.) from 2004 to 2015. Similarly, the massive success of *The Hunger Games* series from 2010 to 2012 created a big unexpected

Figure 3. (Color online) Google Trends for Books from Different Subgenres of Adult Fiction



shock in the demand for children's fantasy books and for juvenile fiction books in general (Roback 2013). As these shocks coincide with our posttreatment period (2008 onward), to mitigate such potential endogenous timing issues, we exclude all romance and juvenile fiction books from our baseline analysis for cleaner identification, although the results are very similar if we include these books as well. Note that these temporal changes in genre demand preferences were not considered in the book choice model of Li (2021). We further discuss some other endogenous timing issues and alternative explanations in Section 6.

4. Data

We acquire data from Nielsen/NPD BookScan on almost all of the books published in print format and sold in the United States (both online and at physical bookstores) between 2004 and 2016 across a wide range of genres, like adult fiction, adult nonfiction, juvenile fiction, and juvenile nonfiction. The BookScan data are considered a gold standard for the measurement of print book sales, and even Amazon gives access to BookScan data to publishers and authors so that they can track their book sales. The data include information on the title, yearly print unit sales, format, price at the time of release, Book Industry Standards and Communications (BISAC) code (provides genre and subgenre information), publisher, and imprint among several other variables. Each genre comprises several subgenres (e.g., romance, thriller, and fantasy are a few of the subgenres of adult fiction; animals, art, and history are a few subgenres of juvenile nonfiction). Based on the first six characters of the BISAC codes, there are over 2,500 such subgenres across different genres. To provide a couple of examples, books like *First Animal Encyclopedia* and *Rain Forest Scratch and Sketch* belong to the juvenile nonfiction genre. Note that we do not have access to any title-level e-book sales data but only have yearly e-book sales data at the genre level, which we utilize in Section 5.5. We conduct our analysis using books published between 2004 and 2015 so that we observe all of the sales in the first two calendar years of a book. By considering the first two calendar years, we are able to include at least the first year of sales of any book, especially of those published in the latter part of a calendar year. This ensures that we capture a big part of sales (including the sales of the paperback version, which is usually released a few months later) for every book because typically, the majority of sales take place during the first year of release. Nevertheless, we also conducted our analysis using sales from only the first calendar year, which yields similar results.

We use adult fiction and juvenile nonfiction books for our baseline analysis in Section 5, and we also analyze other genres, like juvenile fiction and religion, in Section 5.2. We aggregate the data at the book level (each book is identified by a unique combination of title, authors, and publisher) and measure print unit sales for each book within the first two calendar years of its publication.[5] Because we only consider the first two calendar years of release, note that sales are not affected by book vintage. We also restrict our analysis to books that sold more than 100 units in their first two years in the United States so that small absolute changes in unit sales do not lead to great relative changes. Our analysis thus excludes highly unpopular books that sell less than one copy per state per year. About 44% of the books get dropped. The results, however, remain consistent, even if we only exclude those books that sold less than two copies per year in the whole country during their first two years following their release (see Table A.2 in the Online Appendix). We note that a large number of books have near-zero sales, and our objective in this paper is not to study the effects on highly unpopular books with minuscule sales.

Table 1 presents summary statistics of our final sample for different genres and subgenres before (Pre) and since 2008 (Post). There are two things to note here. First, we observe thousands of books published both before and since 2008 across all genres, which provides us with enough statistical power. The number of books published since 2008 is higher because of four more years of data. We further note that there are almost three times more books published in adult fiction than in juvenile nonfiction. We account for this fact in Section 6.1. Second, apart from adult fiction and nonfiction, no other genre or subgenre witnessed a dip in print sales after 2008. As a preview of our results, this provides model-free evidence suggesting a decrease in adult fiction books because of digitization. Further, note that the romance genre did not experience any decrease in sales despite being the most popular genre in the digital format, with e-books accounting for 60% of total unit sales.

Table 1. Count and Average Unit Print Sales Summary

	Number of books		Sales per book	
Genre/Subgenre	2004–2007	2008–2015	2004–2007	2008–2015
Adult fiction	28,133	62,760	8,807 (40,383)	6,926 (44,665)
Adult nonfiction	124,215	222,687	3,611 (26,994)	3,022 (19,761)
Juvenile fiction	17,427	41,661	8,748 (90,898)	9,625 (55,258)
Juvenile nonfiction	9,137	18,097	4,189 (15,312)	4,700 (20,886)
Romance	10,586	23,102	7,260 (17,161)	7,774 (62,578)
Religion	11,659	25,017	2,249 (15,908)	2,616 (21,189)

Note. Standard deviations are in parentheses.

This could be largely because of the huge success of FSG in 2012, which led to a 35% increase in print sales within this subgenre compared with 2011 (Habash 2013).

5. Analysis

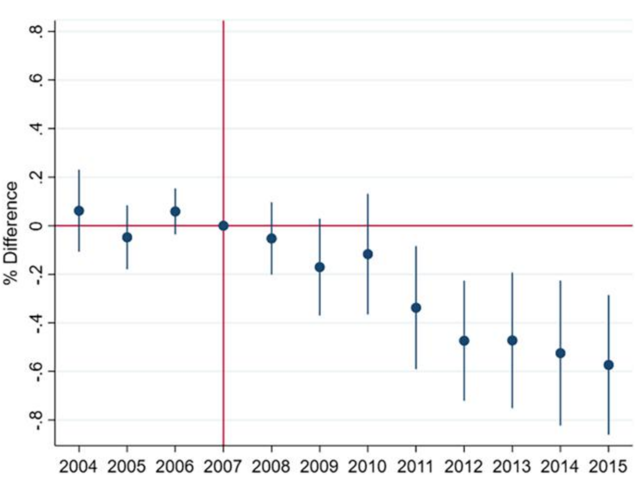
A necessary condition for our DiD approach to give us a causal estimate is that conditional on other covariates, print sales of adult fiction books should have trended similarly to sales of juvenile nonfiction books in the absence of e-readers (i.e., before 2008). To check if the parallel trends assumption holds, we provide a statistical test by estimating the following regression model:

$$\ln(Sales_{ijt,t+1}) = \beta_0 + \boldsymbol{\beta_1\gamma_t} + \boldsymbol{\beta_2\gamma_t} \times AF_i + \beta_3 AF_i + Pub_j + \epsilon_{ijt}, \quad (1)$$

where $\ln(Sales_{ijt,t+1})$ is the natural logarithm of print unit sales in the first two years of book i published in year t by publisher j; $\boldsymbol{\gamma_t}$ is a vector of dummy variables for each publication year (year fixed effects), with 2007 being the base year; and AF_i is a dummy variable for genre, which equals one if book i belongs to the adult fiction genre and otherwise, equals zero (in this section, it takes the value 0 if the genre is juvenile nonfiction, and in Section 5.2, we use other control groups). In addition, we absorb any unobserved heterogeneity across publishers by including publisher fixed effects Pub_j. Note that we do not include book-level fixed effects as our analysis uses one observation for each book (i.e., each book in our sample has a unique publication year). Finally, we cluster the standard errors at the subgenre level to avoid incorrect inference in the difference-in-differences model. Here, β_2 represents the difference in time trends of average sales between AF and JNF books.

We plot the estimate of β_2 (with a 95% confidence interval) for each publication year in Figure 4. As one can see, before 2008, the AF sales trend appears to closely follow the trend of JNF sales, validating the parallel trends assumption. Thus, juvenile nonfiction books seem to be a proper control group to measure the counterfactual print sales of adult fiction print sales after 2007.

Figure 4. (Color online) Unit Sales Difference in Trends Between AF and JNF over Years (Base Year: 2007)



Note that the coefficient estimates are negative from 2008 onward and increase significantly after 2010 when the iPad was launched, indicating a huge and gradual negative effect of e-books on print sales coinciding with the development of the e-book industry. We now estimate the average treatment effect on the treated using the same regression model as above but replace $\boldsymbol{\gamma_t}$ by a binary variable $Post_t$ in the interaction term, which equals zero if book i was published before 2008 and otherwise, equals one. In this model, the estimate of β_2 identifies the effect of digitization on AF print sales:

$$\ln(Sales_{ijt,t+1}) = \beta_0 + \boldsymbol{\beta_1\gamma_t} + \beta_2 Post_t \times AF_i + \beta_3 AF_i + Pub_j + \epsilon_{ijt}. \quad (2)$$

The first column in Table 2 presents the estimates for Model (2). The DiD coefficient is negative and statistically significant, suggesting a 30% ($= 1 - e^{-0.36}$) decrease in print unit sales of AF books on average since 2008. Figure 4 suggests that taking 2011 as the first treatment year would yield a much larger negative effect, mainly because the e-book industry was not as mature before 2011. Indeed, the timing of the propagation of the effect (i.e., increasing effect as e-books gain popularity) gives additional support to our main story.[6] As we explained before, this could also explain why Chen et al. (2019), who analyzed hardcover sales data from 2010 when the e-book market was in its infancy, did not find an effect of delaying the release of books in the digital format.

We also attempt to compare our estimates with those of Li (2021), who analyzed data from 2008 to 2012. She finds that 71.5% of e-book sales come from cannibalizing print books for all of the genres together. Because of

Table 2. Effect Estimates for Adult Fiction vs. Juvenile Nonfiction

	Dependent variable: ln(*Sales*)			
Coefficient	(1)	(2)	(3)	(4)
AF	0.424*	0.484**	0.556**	0.388
	(0.166)	(0.149)	(0.169)	(0.227)
AF × *Post*	−0.360**	−0.337**	−0.307*	−0.423**
	(0.122)	(0.116)	(0.130)	(0.136)
With covariates	No	Yes	Yes	Yes
Publishers	All	All	Major	Nonmajor
N	117,582	117,582	69,246	48,336

Notes. Cluster-robust standard errors (at the subgenre level) are shown in parentheses. All specifications include year and publisher fixed effects. Significance levels are indicated with asterisks.
*$p < 0.05$; **$p < 0.01$.

very different data and approaches, we cannot directly do the comparison. However, using the adult fiction e-book sales (263 million) and print sales (505 million) available for the years 2010–2012 and our cannibalization estimate for pre-2012 (20%), we find that 47.9% of e-book sales came from substituting adult fiction print books.[7] The cannibalization rate for other genres that are less popular in the digital format should be much lower. Based on this back-of-the-envelope calculation, compared with the results reported in Li (2021), which may be invalid because of the data issues that we discussed before, our estimates suggest a lower cannibalization rate of print books for the most popular genre in the digital format.

There are a few confounders that could bias the estimates presented above. For example, if the relative price of adult fiction books with respect to juvenile nonfiction books increased over the years, then this could also explain the decline in adult fiction print sales. To account for the potential confounder, we control for the list price of the books.[8] We find that the estimate remains similar even after controlling for the weighted list price across different print formats (see the second column in Table 2). Note that we do not observe actual selling prices.[9] Because the list price is the maximum price that consumers can pay (and we already control for it), an alternative explanation (based on actual prices) would require that discounts on the list price significantly decreased (or the actual selling price of print books significantly increased) between the years 2010 and 2012 (the period where we see a maximum change in effect size) for adult fiction but not as much for juvenile nonfiction. In column (2) in Table 2, we further control for the publication calendar month and author experience by the number of books. This ensures that the effect is not driven by temporal changes in publication seasonality or author experience between the two genres.

Overall, our results are consistent with what is found in some of the studies that have examined the interaction between physical and digital channels in other information goods industries. For example, Zentner (2006) finds that digital file sharing displaces legal music sales, Ma et al. (2014) find prerelease digital piracy to negatively affect box office revenue, and Gentzkow (2007) finds news websites to cannibalize print readership. We find that the availability of e-books decreases print book sales.

5.1. Major vs. Nonmajor Publishers

We also note that there has been a surge in the number of self-published books even in the print format since 2008 (Statista 2019), especially in the adult fiction genre (Smith 2021). This creates a possibility of the average effect being driven by the lower quality of the additional self-published adult fiction books. To check if this is indeed the case, we estimate the model by only considering books by big publishers that consistently publish a lot of successful books.[10] We find that there is only a modest decrease in the effect size when we only take the big publishers into account (see column (3) in Table 2). As the titles that are published by major publishers are typically not print on demand, the analysis with only major publishers also addresses the concern that our main results could be explained by reduced print inventory because of a rising trend of print-on-demand books. In column (4) in Table 2, we report the results for books published by smaller publishers (including self-published). We find that such books were impacted more negatively. We further analyze the role of growth in the self-publishing of adult fiction books in Section 6.1.

We note that our causal claims could be questioned if an unobservable coincides with the timing of digitization and that also affects adult fiction print sales negatively, juvenile nonfiction print sales positively, or both. For this reason, in what follows, we try to rule out certain factors that could bias our estimates.

5.2. Alternative Genres as the Control Group

In the analysis above, we only considered juvenile nonfiction books as the control group because this genre is expected in theory to be the least affected by the introduction of digital books (as also confirmed by e-book sales patterns in Figures 1 and 2) and shows little evidence for any difference in predigitization trend (see Figure 4). Here, we estimate the effects using print book sales of other genres, such as juvenile fiction, religion, and adult nonfiction, as control groups that are also expected to be significantly less popular than adult fiction in the e-book format (see Figure 1).

In Figure A.2 in the Online Appendix, we compare the pre- and posttreatment sales trends by presenting the estimates of Model (1) using juvenile fiction (Figure A.2(a) in the Online Appendix), religion (Figure A.2(b) in the Online Appendix), and adult nonfiction (Figure A.2(c) in the Online Appendix) as control groups. As one can see from Figure A.2(a) in the Online Appendix, juvenile fiction's pre-2008 trends match adult fiction's trends almost perfectly and then start diverging from 2008 onward. Although the difference in pretreatment trends with religion or adult nonfiction as the control group is also small in size, it is statistically significant. Nevertheless, we again observe a huge divergence in sales trends after 2008 between adult fiction and both religion and adult nonfiction.

In Table 3, we present the estimates of Model (2) with different genres as the control group to measure the average treatment effect on the treated. The first column in Table 3 replicates the results from the first column in Table 2 for reference. We find that the coefficient estimate of $AF \times Post$ is highly negative and statistically significant in all of the cases. As expected, the effect sizes

Table 3. Effect Estimates for Adult Fiction vs. Other Genres

	Dependent variable: ln(*Sales*)			
Coefficient	JNF	JF	REL	ANF
AF	0.424*	0.332**	0.551***	0.198
	(0.166)	(0.108)	(0.127)	(0.110)
AF × *Post*	−0.360**	−0.306*	−0.487***	−0.276*
	(0.122)	(0.127)	(0.111)	(0.115)
N	117,582	149,400	126,922	472,499

Notes. Cluster-robust standard errors (at the subgenre level) are shown in parentheses. All specifications include year and publisher fixed effects. Significance levels are indicated with asterisks.
$^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

with juvenile fiction (JF in the second column in Table 3) and adult nonfiction (ANF in the fourth column in Table 3) are 26% and 24%, respectively, which are somewhat smaller than the effect size with juvenile nonfiction (the first column in Table 3). This is because of the relatively high popularity of e-books in juvenile fiction and adult nonfiction compared with juvenile nonfiction (see Figure 2). Although the effect size with religion books as the control group (REL in the third column in Table 3) is higher (38%) than with juvenile nonfiction, it is partly driven by the positive difference in the pretreatment sales trends. In summary, we find that our main result of about 30% decrease in print sales of adult fiction books is on the same order of magnitude even when we compare adult fiction with other digitally less popular genres.

5.3. Accounting for Publisher Strategies

Since the introduction of e-books, major publishers have adopted different strategies over the years to protect print sales from being cannibalized by e-books. For example, (i) some publishers delayed the release of e-books in 2010; (ii) just before the launch of the iPad in April 2010, Apple and five major publishers, such as HarperCollins, Penguin, and Simon & Schuster, negotiated contracts that allowed each publisher to set retail prices for newly released e-books and the bestseller e-books subject to price caps; or (iii) in October 2014, Amazon and Simon & Schuster agreed to a new multiyear contract under which Simon & Schuster sets the retail prices for its e-books (Trachtenberg 2014). Therefore, it is possible that our estimates could be driven by these publisher-level temporal shocks.

To account for any time-varying unobserved heterogeneity across publishers, we include publisher-year fixed effects in our main specification. Table 4 reports the estimates, including the baseline results reported before in column (1) for reference. We find that our main effect estimate indeed increases (to 36%) after controlling for the publishers' different strategies to counter the substitution of print books over the years. The results suggest that the strategies of delaying the release and adjusting prices succeeded in limiting print sales cannibalization by e-books because the effect size is somewhat larger when we include publisher-year fixed effects.

Table 4. Effect Estimates for Adult Fiction With and Without Publisher-Year Fixed Effects

	Dependent variable: ln(*Sales*)	
Coefficient	No publisher-year FE	With publisher-year FE
AF	0.424*	0.503**
	(0.166)	(0.178)
AF × *Post*	−0.360**	−0.442**
	(0.122)	(0.139)
N	117,582	116,796

Notes. Cluster-robust standard errors (at the subgenre level) are shown in parentheses. The first specification includes year and publisher fixed effects (FEs) individually, whereas the second specification includes publisher-year FEs. Significance levels are indicated with asterisks.
$^*p < 0.05$; $^{**}p < 0.01$.

5.4. Print Format Heterogeneity

Next, we check if the effect varies between paperback and hardcover formats, the two major print formats. As paperback books are cheaper than hardcover books, one can expect the low-priced e-book format to be a closer substitute for the paperback format, assuming that e-book sales are primarily driven by relatively price-sensitive consumers. Thus, the effect size should be larger for sales in paperback format. From Table 5, we can see that the effect is negative, statistically significant, and much larger for paperback (33%) than hardcover (14%), for which the effect, although negative, is smaller and statistically insignificant at the 5% level. However, when we account for publisher-level temporal shocks, we find that although the effect size increases in both

Table 5. Heterogeneous Effects for Adult Fiction Across Print Formats

	Dependent variable: ln(*Sales*)			
Coefficient	Paperback	Hardcover	Paperback	Hardcover
AF	0.284	0.563***	0.352	0.650***
	(0.198)	(0.0995)	(0.213)	(0.116)
AF × *Post*	−0.400**	−0.153	−0.472**	−0.251*
	(0.127)	(0.107)	(0.151)	(0.113)
Publisher-year FE	No	No	Yes	Yes
N	92,025	34,196	91,340	33,640

Notes. Cluster-robust standard errors (at the subgenre level) are shown in parentheses. All specifications include year and publisher fixed effects (FEs). Significance levels are indicated with asterisks.
$^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

cases, it increases more for hardcover sales, leading to a statistically significant impact of 22% on average.

With regard to different print formats, publishers typically follow a "windowing" strategy where they release high-priced hardcover books months before they start selling the cheaper paperback version to price discriminate (De los Santos and Wildenbeest 2017). This means that paperback sales could decrease not only because of the lower price of e-books or strong preferences for the digital format but also, because of the early release of low-priced e-books. To check whether the effect is solely driven by the delayed release of paperback books or not, we estimate the effect using books that were not released in hardcover format because in these cases, the paperback and e-book formats are typically released simultaneously, and we find estimates that are similar to those that we showed for the paperback format above.

5.5. Using Genre E-Book Sales Share as Treatment

So far, our analyses have been based on predefined treatment and control groups (Chetty et al. 2013, Peukert and Reimers 2022). Alternatively, we can use a continuous measure of treatment exposure by exploiting the variation in genre-level e-book popularity across different genres as measured by the fraction of e-book sales of total sales by genre. Note that this strategy does not require us to define treatment and control groups. To do so, we estimate the following regression model:

$$\ln(Sales_{ijt}) = \beta EbookSalesShare_{jt} + \alpha_j + \gamma_t + \epsilon_{ijt},$$

where subscript ijt denotes book i belonging to genre j published in year t; $EbookSalesShare_{jt}$ denotes the e-book sales share of total sales of the genre j (of book i) in year t; and α and γ denote genre and year fixed effects, respectively. In Table 6, we present the results with and without control variables. Taking the estimate in column (1) in Table 6, we find that a 10% point increase in e-book share leads to an average of a 6.6% $(= (1 - e^{-1.077})/10)$ decline in print unit sales. For example, if we apply this estimate to adult fiction (which has an average e-book share of 39% during 2010–2015), this implies a 26% reduction in print sales,[11] which is similar to the 30% reduction that we found using our first specification (see Table 2 above).

Because this approach does not have predefined treatment and control groups, we can similarly apply the estimates in Table 6 to calculate the effects for other genres as well. For instance, when applied to the adult nonfiction genre, the estimate in column (1) in Table 6 suggests a negative effect of digitization of 6.7% as the average e-book sales share in adult nonfiction is 10.1% during 2010–2015. We discuss this in more detail in Section 5.7.

Table 6. Effect Estimates with E-Book Sales Share as Treatment

	Dependent variable: ln(*Sales*)	
Coefficient	(1)	(2)
EbookShare	−1.077** (0.332)	−1.060** (0.329)
With covariates	No	Yes
N	462,976	462,976

Notes. Cluster-robust standard errors (at the subgenre level) are shown in parentheses. All specifications include year and publisher fixed effects. Significance levels are indicated with asterisks.

$^{**}p < 0.01$.

5.6. Matching at the Subgenre Level

Although we observe reasonable pretreatment parallel trends between our treatment and control groups in our main DiD model analysis, we further check the robustness regarding the comparability of our treatment and control groups. Specifically, in our main model analysis, the adult fiction book sample may not be entirely comparable with that of juvenile nonfiction, which can lead to biased estimates. To address such a potential issue regarding the selection of treatment and control groups, we perform propensity score matching. However, one cannot perform matching at the book level as we do not observe the same book being published both before and after treatment. Therefore, we carry out this analysis after aggregating the data at the subgenre-year level as the sales of a subgenre can be observed both before and after digitization. We note that there are thousands of subgenres based on BISAC codes.

Specifically, we redefine the treatment as a binary indicator that equals one if a subgenre belongs to adult fiction and otherwise, equals zero. We first compute the propensity score $p(X_n)$ for each observation n using a logistic regression model. This model includes the average price and the average pretreatment period sales as the matching variables (X_n). After finding the propensity score of each observation, we find a control observation from all observations of other subgenres with the most similar propensity score for each observation in the treatment group.

After this matching process, we check whether the treatment and control groups have similar characteristics. The results of this matching are reported in Table A.5 in the Online Appendix. The two groups appear to be fairly comparable after propensity score matching. Next, we run our main regression based on the matched sample. The first column in Table 7 reports the results using all other subgenres (i.e., all subgenres that do not belong to adult fiction) as the control group, whereas the second column in Table 7 reports the results using only the matched subgenre pairs.[12] We first observe that our primary findings at the book level remain robust as the effect estimate at the subgenre-year level (without matching) is 29%, which is similar to what we

Table 7. Subgenre-Level Analysis with Propensity Score Matching

	Dependent variable: ln(*Sales*)	
Coefficient	Without matching	With matching
AF	0.184** (0.087)	−0.146 (0.148)
AF × *Post*	−0.337*** (0.107)	−0.305* (0.183)
N	22,559	1,296

Notes. Robust standard errors are shown in parentheses. Thes specifications include year and publisher fixed effects. Significance levels are indicated with asterisks.
$^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

obtained in Table 3. Importantly, comparing the estimates in the columns in Table 7, the effect size after matching is only slightly smaller (27%) than the one obtained without matching and statistically significant at the 10% level (because of the low statistical power owing to orders of magnitude fewer observations in the matched sample).

5.7. Effect for Genres Other Than Adult Fiction

Note that the impact of digitization can vary across different genres because of their different exposure to competition from e-books. For this reason, the size of the effect may change based on the definition of the treatment group. In our main analysis, we use adult fiction as the treatment group because theoretically, it is expected to be the most impacted genre. In this subsection, we estimate the effects using other genres, such as adult nonfiction and juvenile fiction, as the treatment group while keeping juvenile nonfiction as the control group. This set of analyses helps us understand how the impact of digitization on print sales varies across different genres. Table 8 reports the results. We find that unlike adult fiction, which experiences a significant negative impact on print sales, the effect estimates for both juvenile fiction and adult nonfiction are smaller; the effect estimates are similar to the 7% reduction that we find in Section 5.5 for the adult nonfiction genre. Moreover, the standard errors are relatively large. In sum, the results imply that our main estimates (for adult fiction) are larger than for other genres. It is also to be noted that, theoretically, one would indeed expect the effect of digitization on print sales to be smaller for genres other than adult fiction because of their lesser popularity in the e-book format.

Table 8. Effect Estimates for Different Genres

	Dependent variable: ln(*Sales*)		
Coefficient	AF	JF	ANF
Treated	0.424* (0.166)	0.157 (0.114)	0.195 (0.111)
Treated × *Post*	−0.360** (0.122)	−0.081 (0.077)	−0.079 (0.061)
N	117,582	85,977	409,010

Notes. Cluster-robust standard errors (at the subgenre level) are shown in parentheses. All specifications include year and publisher fixed effects. Significance levels are indicated with asterisks.
$^*p < 0.05$; $^{**}p < 0.01$.

6. Alternative Explanations

In this section, we carry out tests to check how certain supply-side or demand-side factors that coincide with the timing of digitization affect our results. First, we check if the main results continue to hold if we restrict the genres in treatment and control groups to have the same number of books and the same percentage of their distributions. Second, we investigate if our results could be explained by the rise of other internet media, possibly lowering the demand for book reading in general. Last, we discuss how advertising variation across genres could potentially influence our estimates.

6.1. Variation in the Number of Books Across Genres

We showed before that each year, more adult fiction books are published than juvenile nonfiction books (see Table 2).[13] Further, this difference has increased significantly over the years, especially if we include titles that sell very few copies. As digitization has reduced the costs of bringing new products to market (Waldfogel 2017), there has been a rise in self-publishing, particularly in the fiction category. Thus, the average number of books published per year increased by 67% between 2004–2007 and 2008–2015 in adult fiction, whereas it only increased by 15% in the juvenile nonfiction genre.

To ensure that the rising difference in the number of books published between the two genres is not driving our results, we consider the same number of books from both genres every year. We do the analysis with only the top 100/500/1,000/2,000 books by sales per year from each of the two genres. We present the estimates of Model (2) in Table 9. We continue to find that the effect is negative and statistically significant irrespective of how many top books we consider. We also note that the effect size is slightly smaller for more popular books than the effect of 30% from Table 2 using the overall sample, and it increases as we include more books from each genre.

Note that if the overall demand for print books in each genre remained constant over the years, then the decline in average print unit sales of adult fiction books after 2007 could also be a result of an increase in the number of books released in the adult fiction genre relative to other genres (i.e., because of relatively higher level of competition in adult fiction (the demand for

Table 9. Effect Estimates for the Same Number of Top Books

Coefficient	Top 100	Top 500	Top 1,000	Top 2,000
AF	1.529***	1.513***	1.718***	2.080***
	(0.0696)	(0.0902)	(0.111)	(0.135)
AF × *Post*	−0.185**	−0.210**	−0.222**	−0.300***
	(0.0647)	(0.0651)	(0.0670)	(0.0713)
N	2,378	11,957	23,919	47,849

Notes. Cluster-robust standard errors (at the subgenre level) are shown in parentheses. All specifications include year and publisher fixed effects. Significance levels are indicated with asterisks.
$^{**}p < 0.01$; $^{***}p < 0.001$.

adult fiction books splits among a higher number of choices)). However, best-selling adult fiction books are unlikely to face competition from long-tail books. Liebowitz et al. (2021) find that the increase in the number of books available had almost no impact on the sales of top books. Their data show that the top-ranked books continued to account for a very similar fraction of the total print sales over the years. We show that a similar result holds for the adult fiction genre in Figure A.3 in the Online Appendix. For instance, the top 500 titles in the adult fiction genre take about 65% of all sales in this genre. We can see in the first column in Table 9 that even the top-selling adult fiction books experience a significant decline in print sales compared with the best-selling juvenile nonfiction books, suggesting that demand diversification among the increased alternatives is unlikely to explain our results.

We further note that our main analysis does not necessarily imply that overall AF print sales also declined. This is because having digital versions may have reduced the sales of each print book title, but there are also more print book titles over time (a fallacy of the composition). As the top 100/500/1,000/2,000 books account for a different share of sales in each genre every year, the analysis in Table 9 would have the same limitation. Therefore, in Table 10, instead of using the top absolute number of books, we use the sample of the top relative number of books by considering the top percentiles each year in each of the two genres. We continue to find a negative impact of digitization on print book sales of adult fiction.[14] These results suggest that our main conclusions are robust to potential digitization effects on the supply of new titles or on the distribution of sales across genres. Moreover, we show in Section 6.2 that overall print sales indeed declined after digitization for adult fiction.

6.2. Potential Changes in the Demand for Fiction Books

So far, we have established that adult fiction print book sales decreased significantly since 2008 compared with other genres that are less popular in the digital format. However, this decrease in book sales may be driven by (i) people shifting to other forms of media, particularly online music and video platforms like YouTube (Dukper et al. 2018), and/or (ii) a lower quality of books published after 2008 in the adult fiction genre rather than because of cannibalization. To check if these factors are driving the results, it would be useful to see how the overall (print and e-book) sales have changed over the years. Unfortunately, we do not have e-book sales data at the book level for the books in our sample to see the evolution of total sales. Nevertheless, we obtained aggregate sales data from Nielsen/NPD at the genre level for print and e-book formats. In Figure 5, we plot the total yearly sales of adult fiction books from 2004 to 2007 and from 2010 to 2013.[15] One can see that the total sales have actually increased after 2008, suggesting there was no decline in the overall demand for adult fiction books as would need to be the case under the alternative explanations (i) and/or (ii). One of the reasons behind the rise in overall sales could be that e-books allow people to read more books. For example, while traveling, travelers cannot bring 20 print books with them, but they can read them on a device (Smith and Zentner 2016). Similarly, e-readers avoid some privacy concerns because they do not show the book cover when reading adult fiction books in a public place.

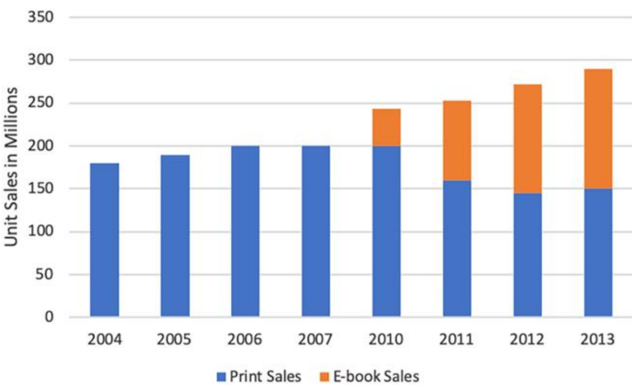
At the same time, the growth in e-book sales seems to be in tandem with the decline in print sales, strongly indicating cannibalization. Furthermore, from Figure 5, it is clear that the total print unit sales of adult fiction

Table 10. Effect Estimates for Top-Selling Books

Coefficient	Top 2nd percentile	Top 5th percentile	Top 20th percentile	Top 50th percentile
AF	0.813***	0.554***	0.502***	0.529***
	(0.0510)	(0.0641)	(0.0776)	(0.112)
AF × *Post*	−0.389***	−0.416***	−0.403***	−0.441***
	(0.0686)	(0.0632)	(0.0654)	(0.0845)
N	2,336	11,774	23,558	58,911

Notes. Cluster-robust standard errors (at the subgenre level) are shown in parentheses. All specifications include year and publisher fixed effects. Significance levels are indicated with asterisks.
$^{***}p < 0.001$.

Figure 5. (Color online) Total Sales by Year for Adult Fiction



Source. Adapted from Nowell (2015).

books were not constant but declined over the years. This gives us more confidence that the decrease in average print sales of adult fiction books cannot be entirely attributed to an increase in the supply of books (i.e., the fallacy of the composition).

6.3. Role of Advertising

Another potential concern with our identification strategy would be that temporal shocks to advertising at the genre-format level could also explain our results—specifically, if advertising for juvenile nonfiction print books and/or adult fiction e-books increased after digitization. We reached out to industry experts and researchers who have conducted research on the book industry, and according to them, advertising is concentrated on top titles; long-tail titles have limited or no advertising. Hence, we conduct analyses that show that the effects are similar for top-selling versus nontop-selling titles (see Table A.6 in the Online Appendix). Because top-selling and nontop-selling titles likely have different advertising expenditures and yet, the results across top-selling and nontop-selling titles are similar, these results suggest that advertising variation is unlikely to substantially affect our estimates.[16]

7. Discussion and Conclusions

In this paper, we study whether and by how much digitization affected the print sales of adult fiction books. To estimate the impact, we use the print sales of juvenile nonfiction books, which are in very low demand in the e-book format, to provide us with the counterfactual print sales trends. Using print sales data on a large fraction of all of the books published and sold in the United States between 2004 and 2015 as well as difference-in-differences and other identification strategies, we find that, since 2008, print sales of adult fiction books decreased by about 24%–38%. Importantly, the effect remains high (>20%) for all levels of popularity (see Table 9) and is driven primarily by the loss in paperback sales rather than in hardcover sales (see Table 5). In Table A.7 in the Online Appendix, we summarize the estimates obtained by using different empirical specifications. In Table A.8 in the Online Appendix, we also provide a summary of heterogeneous effect estimates (using the specification in the first row of Table A.7 in the Online Appendix) for different subsamples. These heterogeneous effects range from 18% to 41%. Moreover, we find that the effect estimates are much smaller (about 8%) for the adult nonfiction and juvenile fiction genres, likely because of their lesser popularity in the e-book format.

Although it is possible that our results are partly driven by the illegal sharing of e-books (digital piracy) rather than legal sales of e-books, this is unlikely because digitally pirated copies mostly affect the sales of e-books as physical versions are not as close a substitute for the free pirated electronic version (see Reimers 2016). In addition, we cannot make strong claims regarding the effect on publishers' revenue or profits from combined sales of print and digital books. As we do not have title-level e-book sales data, we also cannot be sure if the substitution was within title or across titles, which might lead to different implications for publishers. Nevertheless, given that we see a significant reduction in print sales for even very popular books (see Table 9), which are not easily substitutable by other book titles and also very likely to have an e-book version available, this suggests that the results are at least in part driven by within-title substitution.

Our paper has important implications for managers in the publishing industry. The results strongly suggest that the digital channel reduces adult fiction book sales in the print format on the order of 24%–38%. This is important information that publishers should consider when thinking about their release timing and pricing strategies. Like in the movie industry, where producers delay the availability of a movie on DVD or the online channel in response to the cannibalization of box office sales by DVDs or online streaming platforms, the book publishers can also adopt a similar windowing strategy for the release of e-book version. In fact, our analysis in Section 5.3 indicates that the strategies of delaying the release and adjusting prices indeed mitigated the cannibalization of print books for the publishers. This result contrasts with the recommendations of Chen et al. (2019), who argue that delaying the release of books in the digital format did not affect sales of hardcover books. The difference can be explained by the time period of the analysis in the work by Chen et al. (2019), who analyzed data from 2010 when e-book adoption was a nascent market. Publishers may also seek to lower the price difference between e-book and print versions, especially compared with the paperback version, which seems to be cannibalized at a higher rate than the hardcover format.

We also believe that our results provide important and scarcely available empirical evidence that can inform the legal debate concerning how the mass third-party digitization and free provision of digital books affect physical sales (e.g., U.S. Copyright Office Fair Use Summary: Authors Guild, Inc. v. Google, Inc. 2015). The theory underlying this debate is that mass digitization and free access foster discovery, increasing the demand for physical books, but mass digitization and free access also provide a zero-cost substitute, decreasing the demand for physical books. Although our context does not exactly match the context of the legal debate because e-books are sold at positive prices, e-books do provide a lower-cost alternative that would in theory lead to both substitution and discovery. Our results in the context of positive prices suggest that the substitution effect is greater than the product discovery effect and that the book digitization net impact of these two forces is to substantially decrease the demand for physical sales. Regarding this debate, our study complements the results in a related work by Nagaraj and Reimers (2023), whose sample of books includes old and not currently popular titles, whereas our sample is modern and includes currently popular books. One possibility is that substitution dominates search and discovery effects for current books, whereas search and discovery dominate substitution for older and more obscure books, such as those in the sample used in Nagaraj and Reimers (2023). In this way, our results and those in Nagaraj and Reimers (2023) can provide clues about how substitution and discovery balance out across the book's vintage dimension.[17] Relatedly, in the context of the Google Books platform, which only shows a "snippet" view (i.e., preview) of new books, the substitution mechanism is muted, and the platform then is primarily used for searching/discovering a book. On the other hand, our context is about buying an e-book as an alternate channel for reading.

Our work is not without limitations. First, we acknowledge that for our causal identification strategy, it would have been preferable to have exogenous variation in e-reader adoption across individuals or in e-book format availability across book titles. Second, because of the absence of e-book data on price, we are unable to control for e-book prices. Not observing e-book prices and sales also prevents us from disentangling the roles of price difference between print books and e-books and of consumers' preferences regarding the format behind our results. Using data on e-book sales and prices may also enable one to study market expansion, which is an avenue for future research perhaps using data from a recently created company Bookstat, which provides data on sales of e-books.[18] There is also a concern about the potential violation of the stable unit treatment value assumption as parents with e-readers might create a positive spillover effect on the demand for juvenile e-books. We note that this interference would only lead our estimates to be smaller (i.e., closer to zero) of the effect of digitization on adult fiction print books because it should make the demand for print books in the two genres more similar, biasing our estimated effects toward zero. However, the relationship rests on assuming that the impact of digitization on print sales is also negative for the control group (juvenile nonfiction in our case). Another limitation is that we only have four years (2004–2007) of pretreatment data to check parallel trends between the treatment and control groups. However, the difference in sales is very small and statistically insignificant in 2008 as well (see Figure 4) when the adoption of digital readers was at a very nascent stage, giving us more confidence in our parallel trends assumption.

Endnotes

[1] E-readers and tablets are the two most popular devices on which to read e-books (Greenfield 2013). Our definition of digitization embraces multiple "digitization" factors as is usual in the digitization literature, including for example, a large number of studies on digital piracy. In our setting, these factors include the release of e-reading devices, the release of books in digital format, the development of platforms that sell books in digital formats, or the launch and development of self-publishing platforms.

[2] We can speculate that Esri projects its data by analyzing purchases of books by characteristics at the national level and then using the demographic composition of each county to split national purchases across counties. However, if done this way, it would be difficult to then use geographic or time variation because all of the data variation would come from a single national data point for each time that the survey is conducted.

[3] Variable QBR3MCX in the Consumer Expenditure Survey contains information on book sales but does not separate e-book from print book sales. This variable is only available for a few of the largest MSAs.

[4] Suppose the fraction of e-book sales increases from 0% to 50%. Note that this does not necessarily imply that sales of "print" books decrease or increase; it only means that "e-book" sales increased from zero to a positive amount during the study period.

[5] This implies that we also include sales from 2008 (a posttreatment year) for the books published in 2007 (a pretreatment year). However, we note that the effect of digitization in 2008 would be very small because of the very low adoption of e-readers. Even if there was a negative effect in 2008, this would make our overall effect estimates smaller.

[6] See Table A.3 in the Online Appendix for the heterogeneous effect estimates over different periods. We find that the effect is larger (41%) during 2012–2015 compared with 2008–2011, during which the effect is smaller and statistically insignificant.

[7] The calculation is $\left(\frac{505}{1-0.2} - 505\right)/263 = 0.479$.

[8] Note that a book title can be sold in multiple print formats (e.g., hardcover and paperback). We weight list prices with sales across formats.

[9] The list price of print books is typically printed on the cover, making it unlikely that print books sell at a higher price than the list price as doing so might anger consumers. In fact, the printing of prices on books has been portrayed as leading to a book cartel (Nguyen 2021).

[10] We define a major publisher as being in the top two percentile in terms of total revenue and revenue per book. See the summary

statistics for the two groups of publishers in Table A.4 in the Online Appendix. There are 54 major publishers in our sample, and they account for more books than all of the nonmajor publishers.

[11] We use data on genre-level e-book yearly sales from PubTrack Digital, which only started tracking these data from 2010 onward. We were able to obtain these data for the years 2010–2015. As we do not observe data for the years 2008–2009, we can approach our analysis in one of two ways. (i) We drop the two years, or (ii) we assume that the e-book sales share in 2008–2009 is the same as in 2010. We present our results using the first approach. We find that the estimates are only slightly smaller (23%) if we take the second approach.

[12] The number of observations in the second column (1,296) includes 648 observations (subgenre-year) in the treatment (adult fiction subgenres) and another 648 observations in the matched control group (nonadult fiction subgenres).

[13] Before 2008, on average about 7,000 and 2,300 books (selling more than 50 copies per year) are published every year in adult fiction and juvenile nonfiction, respectively. These numbers increase to 18,000 and 7,000, respectively, if we consider all of the titles.

[14] Note that the estimates in the two tables are not comparable because adult fiction has a lot more books than juvenile nonfiction. Specifically, unlike in Table 9, where each genre has an equal number of books, in Table 10, *x* percentile in AF would include far more books than *x* percentile in JNF.

[15] No e-book sales data are available for years before 2010 from Nielsen.

[16] The same experts also revealed that particularly until 2015, advertising in this industry was mainly at the level of the title and not so much at the level of the print format versus the digital format. If so, advertising would only promote the title but not alter consumers' choice of print versus digital format.

[17] Indeed, Nagaraj and Reimers (2023) conduct a heterogeneity analysis and find no statistically significant effects for the most popular books in their sample of old books. Nagaraj and Reimers (2023) also study how the digitization of lesser-known and perhaps forgotten works affects the supply of physical editions of these works.

[18] Bookstat began collecting data in 2016, and therefore, their data do not overlap our study period.

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